

ORF309/MAT380

Basic principles II:

Probabilistic modeling &

Conditional probability (pp.12-19)

Ioannis Akrotirianakis

Agenda for today

- Simple examples of the 3 Axioms
- Probability Modeling
- Discrete/Continuous Models
- Conditional Probability
- Bayes formula

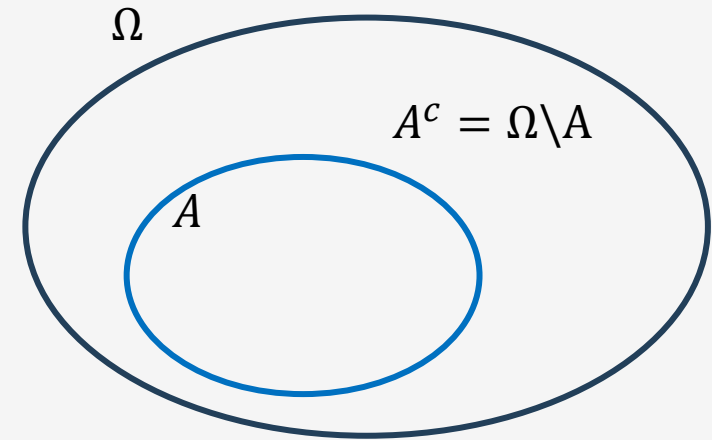
Recap from Lecture #2

- We introduced the basic ingredients that help us model any Random Experiment (RE)
 - **Sample space** (Ω)
 - **Events** ($A, B, C, \dots \subseteq \Omega$)
 - **Probability measure:** $P(A)$ amount of confidence that event A will come up TRUE
 1. $0 \leq P(A) \leq 1$ (degree of confidence)
 2. $P(\Omega) = 1$ (something must happen)
 3. IF $A_i \cap A_j = \emptyset, \forall i \neq j$ then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ (prob. of mutually exclusive events)

3 Axioms by
Kolmogorov – 1933

Simple derivations from the 3 Axioms

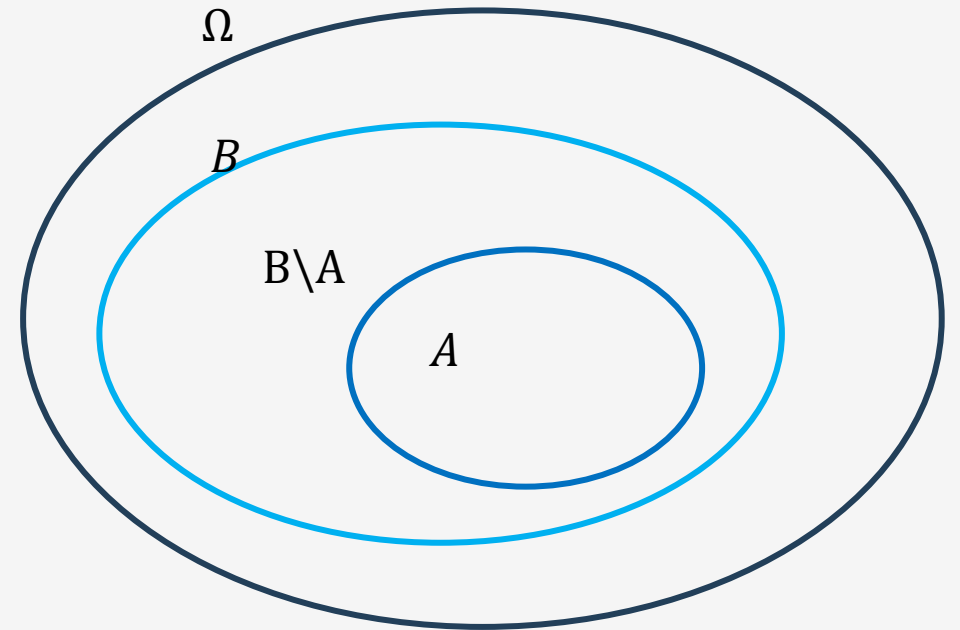
- What is $P(A^c)$ =?
- **CLAIM:** $P(A^c) = 1 - P(A)$, but WHY?



- We can write: $\Omega = A \cup A^c$
- We can also note: $A \cap A^c = \emptyset$ (mutually exclusive)
- Then life is easy by taking probabilities:
 - $P(\Omega) = P(A \cup A^c) \Rightarrow 1 = P(A) + P(A^c) \Rightarrow P(A^c) = 1 - P(A)$

Simple derivations from the 3 Axioms

- Assume you have 2 events A, B
- Assume also that A **implies** B , i.e., $A \subseteq B$
- **CLAIM:** $P(A) \leq P(B)$
 - Write: $B = A \cup (B \setminus A)$ and note that $A \cap (B \setminus A) = \emptyset$
 - Take probabilities: $P(B) = P(A \cup (B \setminus A)) = \underbrace{P(A)}_{\geq 0} + \underbrace{P(B \setminus A)}_{\geq 0} \geq P(A)$

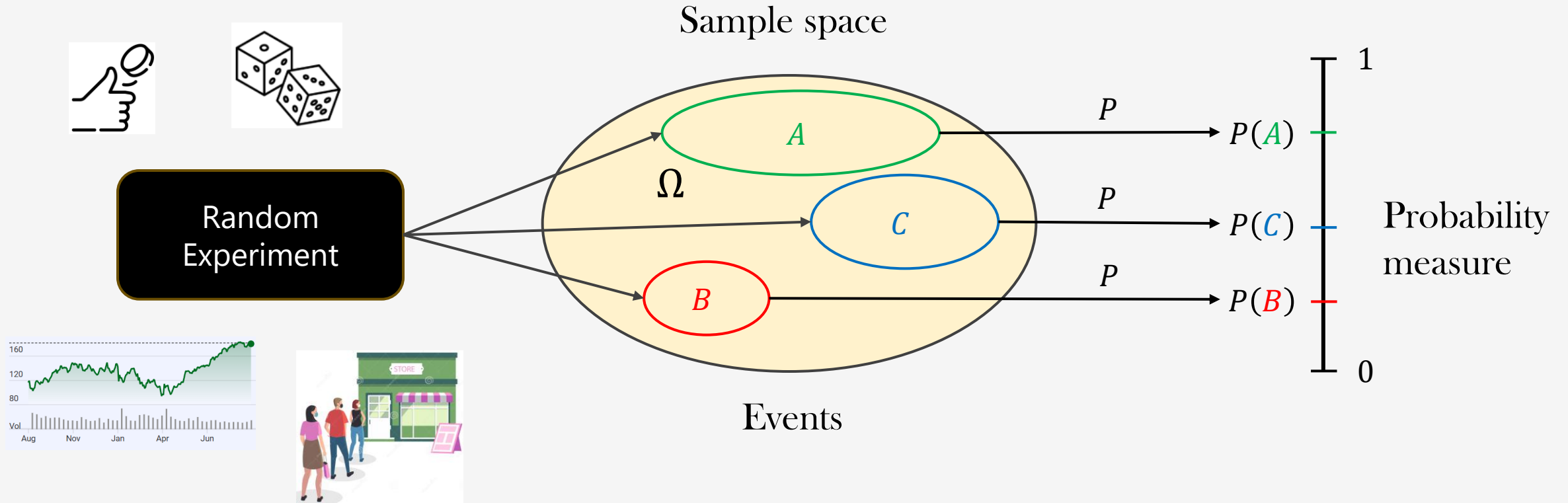


Example:

A = "it rains", B = "It is cloudy". $A \subseteq B \Rightarrow P(A) \leq P(B)$

Probabilistic Modeling

The main elements of a Probabilistic model



Probabilistic Modeling

- **What is it?**
 - It is the structured process (and **art**) of describing a random experiment (RE) in terms of Probability Theory
 - In other words: making the right assumptions about the RE and the events of interest **and** computing (theoretically) the probabilities of these events.
 - **NOTE**: the *empirical computation* of the probabilities is the job of Statisticians

Probabilistic Modeling

- **Example 1:**
 - Random Experiment: throw a die
 - Sample space: $\Omega = \{1,2,3,4,5,6\}$
 - Event: $A = \text{"the die shows an even number"}$ $\rightarrow A = \{2,4,6\}$
 - **What is the probability of event A ??** (Notation: $P(A) = ? ?$)
 - Modeling assumptions
 - Every number is **equally likely** to come up (**fair die** – NO favors!)

Probabilistic Modeling

- Modeling assumptions (let's analyze them – what is the thought process?)
 - Every number is **equally likely** to come up (**fair die** – NO favors!)
 - How can I compute the probabilities ??
 - From the assumptions: $P\{1\} = P\{2\} = \dots = P\{6\}$ (but still I do not know their value (?))

Probabilistic Modeling

- Modeling assumptions (let's analyze them – what is the thought process?)
 - Every number is **equally likely** to come up (**fair die** – NO favors!)
 - How can I compute the probabilities ??
 - From the assumptions: $P\{1\} = P\{2\} = \dots = P\{6\}$ (but still I do not know their value (?))
 - But I can safely say:
 - $\Omega = \{1\} \cup \{2\} \cup \dots \cup \{6\}$ **and** $\{1\} \cap \{2\} \cap \dots \cap \{6\} = \emptyset \Rightarrow$ Mutually exclusive (atomic) events

Probabilistic Modeling

- Modeling assumptions (let's analyze them – what is the thought process?)
 - Every number is **equally likely** to come up (**fair die** – NO favors!)
 - How can I compute the probabilities ??
 - From the assumptions: $P\{1\} = P\{2\} = \dots = P\{6\}$ (but still I do not know their value (?))
 - But I can safely say:

Mutually exclusive (atomic) events

$$\Omega = \{1\} \cup \{2\} \cup \dots \cup \{6\} \quad \textbf{and} \quad \{1\} \cap \{2\} \cap \dots \cap \{6\} = \emptyset \quad \Rightarrow$$

$$\Rightarrow P(\Omega) = P(\{1\} \cup \{2\} \cup \dots \cup \{6\}) \Leftrightarrow 1 = P\{1\} + P\{2\} + \dots + P\{6\} = 6P\{i\} \Leftrightarrow P\{i\} = 1/6$$

for all $i = 1, 2, \dots, 6$

Probabilistic Modeling

- Now we have everything we need to compute $P(A) = P\{2,4,6\} = ?$

- Note that:

Mutually exclusive (atomic) events

- $A = \{2,4,6\} = \{2\} \cup \{4\} \cup \{6\}$ and $\{2\} \cap \{4\} \cap \{6\} = \emptyset$

- Apply the 3 Axioms:

- $P(A) = P(\{2\} \cup \{4\} \cup \{6\}) = P\{2\} + P\{4\} + P\{6\} = 3 \frac{1}{6} = \frac{1}{2}$

- **End of the Example**

structure of probabilistic modeling & reasoning:

assumptions → sample space → atomic events →
→ use the 3 axioms → computation

Discrete Probability Modeling

- **Remark:** In general, if Ω is a finite or countable set (discrete set), then for any event $A \subseteq \Omega$ we can write: $A = \bigcup_{i \in A} \{i\}$ (union of **atomic** events)
 - Since $\{i\} \cap \{j\} = \emptyset, \forall i \neq j$ we can also write
 - $P(A) = P(\bigcup_{i \in A} \{i\}) = \sum_{i \in A} P\{i\} = \frac{|A|}{|\Omega|}$

Because of our **modeling assumption** that each outcome in Ω has **equal probability!**
 - where $|A|$ = "the number of outcomes in the event A " (similarly for $|\Omega|$)
- In the previous example: $P(A) = P(\bigcup_{i \in A} \{i\}) = \frac{|A|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$

Continuous Probability Model

- **EXAMPLE 2**

- Random Experiment: We observe the exact time a bus arrives at the bus stop.

- Modeling assumptions:

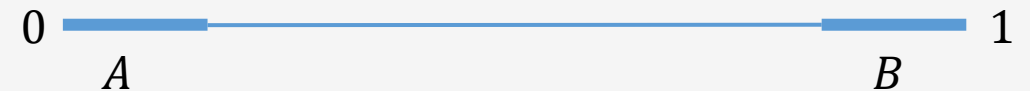
- 1) The bus always arrives in the **1st hour**

- 2) It is equally likely the bus to arrive at any time in that hour.

- What is the sample space?

- $\Omega = [0,1]$ ($\Omega \subset \mathbb{R}$, **not** countable, **not** finite)

- Events: $A = [0.0, 0.1]$ and $B = [0.9, 1.0]$
 $0.1 \rightarrow 10\% \text{ of } 1 \text{ hour} \rightarrow 0.1 \cdot 60 = 6 \text{ minutes}$



Continuous Probability Model

- **EXAMPLE 2**
 - How can we compute the probabilities of A and B ?? Any intuition ??

Continuous Probability Model

- **EXAMPLE 2**

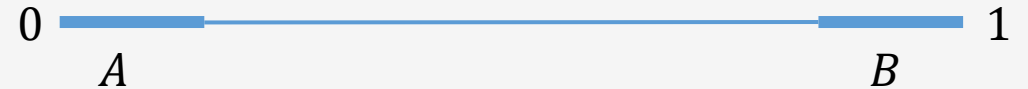
- How can we compute the probabilities of A and B ?? Any intuition ??
- An **intuitive** idea is: $P(A) = \text{length}(A) = \text{length}([0,0.1]) = 0.1$ (or 10% confident that A will happen)
- Let's see where this idea leads us...

- **Check the 3 Axioms:**

1. $0 \leq P(A) \leq 1$ and $0 \leq P(B) \leq 1 \rightarrow \text{OK}$

2. $P(\Omega) = P([0,1]) = 1 \rightarrow \text{OK}$

3. If $A \cap B = \emptyset$ then $P(A \cup B) = P(A) + P(B) \rightarrow \text{OK}$



Continuous Probability Model

- **EXAMPLE 2**
 - Everything ***seems*** to be going alright

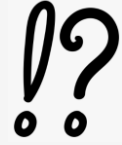
Continuous Probability Model

- **EXAMPLE 2**

- Everything *seems* to be going alright
- **BUT wait a minute!** We still need to compute $P\{x\}$ for a random point $x \in [0,1]$??
 - $\{x\}$ is the event that **the bus arrives exactly at** x
 - $P\{x\}$ is the probability that the bus arrives **exactly** at time x
- Well, let's use the intuitive definition: $P\{x\} = \text{length}(\{x\}) = 0$ **!?**

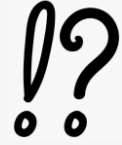
Continuous Probability Model

- **EXAMPLE 2**

- We have just shown the following *paradox*: 
- According to our assumptions, the bus will arrive in the 1st hour
- But the probability of the bus arriving any specific point x within that hour is 0

Continuous Probability Model

- **EXAMPLE 2**

- We have just shown the following **paradox**: 
- According to our assumptions, the bus will arrive in the 1st hour
- But the probability of the bus arriving any specific point x within that hour is **0**
- For **continuous** sample spaces the event $\{x\}$ is just the outcome x
 - **Outcomes of random experiments do not have probability!!**
- We only assign probabilities to events!

Continuous Probability Model

- **EXAMPLE 2**

- **Interpretation of $P\{x\} = 0$ for continuous sample spaces:**

- $P\{[0,0.1]\} = 0.1$, $P\{[0,0.001]\} = 0.001$, $P\{[0,0.00001]\} = 0.00001 \dots$
- As the length of the interval decreases, our **confidence** that the bus will arrive within that interval decreases.
- The probability of being correct **decreases** ($P\{correct\} \rightarrow 0$)
- The probability of being wrong **increases** ($1 - P\{correct\} \rightarrow 1$)
- Hence, we cannot predict something with infinite precision (e.g., Quantum Mechanics)

Continuous Probability Model

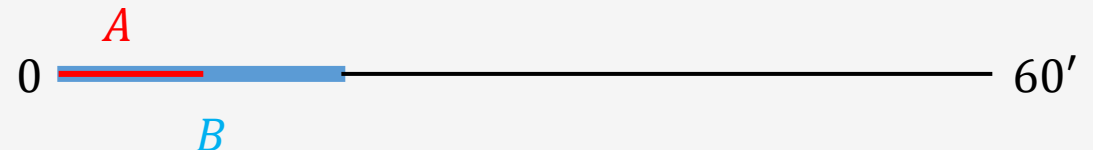
- **Always remember!**
 - We only specify probabilities for events and NOT for outcomes!
 - For Continuous Probability Models where $\Omega \subset \mathbb{R}$ (e.g., $\Omega = [0,1]$) we have:
 - $\overset{\text{event}}{\{x\}} = \overset{\text{outcome}}{x}$ and $P\{x\} = 0$
 - Then we would have $1 = P(\Omega) = P\{\cup_{x \in \Omega} \{x\}\} \neq \sum_{x \in \Omega} P\{x\} = 0$

Wrong because Ω is NOT countable when $\Omega \subset \mathbb{R}$

Conditional Probability

Conditional Probability

- **Let's continue with Example 2:**
 - We wait at the bus stop and the bus comes in the first 60'
 - **Modeling assumption:** The bus is equally likely to arrive at each time
 - Let's define **two events**
 - A = "the bus arrives in the first 10 minutes" $\rightarrow A = [0, 10']$
 - B = "the bus arrives in the first 20 minutes" $\rightarrow B = [0, 20']$
 - **What is $P\{A\}$ = ??**



Conditional Probability

- **Let's continue with Example 2:**

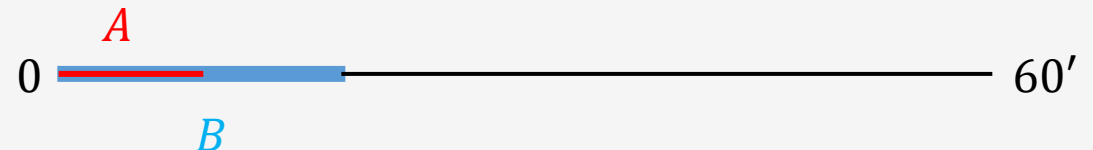
- We wait at the bus stop and the bus comes in the first 60' (1 hour)
- **Modeling assumption:** The bus is equally likely to arrive at each time

- Let's define **two events**

- A = "the bus arrives in the first 10 minutes" $\rightarrow A = [0, 10']$
- B = "the bus arrives in the first 20 minutes" $\rightarrow B = [0, 20']$

- **What is $P\{A\}$ = ??**

- **If you know $B = TRUE$ what is $P\{A\}$ = ??**



Conditional Probability

- **Definition:** Suppose you have two events A, B . The conditional probability of event A given that the event B happened (i.e., $B = \text{TRUE}$) is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Intuition:**
 - Run your random experiment many times and record the results for events A **and** B (you have $A = \text{True/False}$ and $B = \text{True/False}$);
 - **Keep only** experiments where B happened (you now have $A = \text{True/False}$ and $B = \text{True}$);
 - Look at the **fraction** in these experiments that A happened too.

Conditional Probability

- **Example:**

- You have a die and consider the two events $A = \{2\}$ and $B = \{2,4,6\}$.
- What is $P(A|B) = ?$

- **Intuition:**

- Perform the random experiment many times and record outcomes
- Keep the experiments where $B = \{2,4,6\}$; ($B = \text{TRUE}$ and A can be anything)
- Find the **fraction** where $A = \text{TRUE}$

Conditional Probability

- Is $P(\cdot | B)$ a probability??

Conditional Probability

- Is $P(\cdot | B)$ a probability?? → Check if the 3 Axioms hold??

Conditional Probability

- Is $P(\cdot | B)$ a probability?? → Check if the 3 Axioms hold??

1. $0 \leq P(A|B) = \frac{P(A \cap B)}{P(B)} \leq 1,$ Why?

2. $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = ??$

3. If A, A' are 2 mutually exclusive events ($A \cap A' = \emptyset$), then

$$P(A \cup A'|B) = ??$$

Conditional Probability

- Is $P(\cdot | B)$ a probability?? → Check if the 3 Axioms hold??

1. $0 \leq P(A|B) = \frac{P(A \cap B)}{P(B)} \leq 1$, because $P(A \cap B) \leq P(B)$ (by simple logic and $A \cap B \subseteq B$) implies

2. $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} =$??

3. If A, A' are 2 mutually exclusive events ($A \cap A' = \emptyset$), then

$P(A \cup A'|B) =$??

Conditional Probability

- **Is $P(\cdot | B)$ a probability??** → Check if the **3 Axioms** hold??

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Conditional Probability

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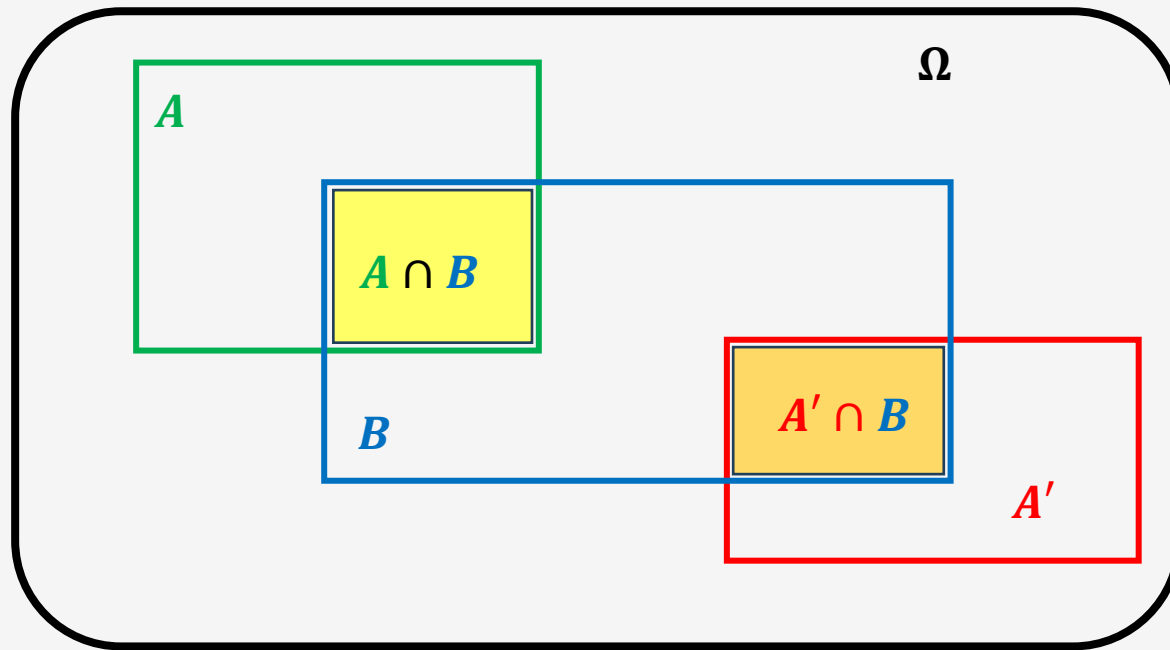
$$\begin{aligned} P(A \cup A'|B) &= \frac{P((A \cup A') \cap B)}{P(B)} = \frac{P((A \cap B) \cup (A' \cap B))}{P(B)} = \frac{P(A \cap B) + P(A' \cap B)}{P(B)} \\ &= P(A|B) + P(A'|B) \end{aligned}$$

Conditional Probability

- **Illustration of 3rd axiom**

$$P(A \cup A'|B) = P(A|B) + P(A'|B)$$

$$A \cap A' = \emptyset$$



Bayes formula

Bayes formula



Rev. Thomas Bayes
1701-1761

- **Connects** $P(A|B)$ **with** $P(B|A)$
 - **Useful when one of them is easier to compute than the other**
- **Example**: A popular COVID test has the following characteristics:
 - If a person is **sick**, the probability of testing positive is = 95%
 - If a person is **not sick**, the probability of testing positive = 2%
 - 1 person in 1000 is sick
- **Question**: What is the probability that a person is sick, when the test of that person comes back positive

Bayes formula

- What are our **Events??**
 - $S = \{sick\}$ and $Q = \{test\ is\ positive\}$
 - What probabilities we know already??
 - $P(S) = \frac{1}{1000}$, $P(Q|S) = 0.95$, $P(Q|S^c) = 0.02$
- Which probability we are looking for??
 - $P(S|Q) = ??$

Bayes formula

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- We will use **Bayes formula** to compute $P(S|Q)$

$$\begin{aligned} P(S|Q) &= \frac{P(S \cap Q)}{P(Q)} \Leftrightarrow P(Q)P(S|Q) = P(S \cap Q) \Leftrightarrow \frac{P(Q)P(S|Q)}{P(S)} = \frac{P(S \cap Q)}{P(S)} \Leftrightarrow \\ &\Leftrightarrow \frac{P(Q)P(S|Q)}{P(S)} = P(Q|S) \Leftrightarrow P(S|Q) = \frac{P(S)P(Q|S)}{P(Q)} \end{aligned} \quad (1)$$

- In the above formula **we know**: $P(S)$ and $P(Q|S)$.
- We only **need to compute** $P(Q)$

Bayes formula

$\Omega = \{\text{all people}\}$

- But I can write: $P(Q) = P\left(Q \cap \overbrace{(S \cup S^c)}^{\Omega = \{\text{all people}\}}\right) = P((Q \cap S) \cup (Q \cap S^c)) =$
 $= P(Q \cap S) + P(Q \cap S^c) = P(Q|S)P(S) + P(Q|S^c)P(S^c) \quad (2)$

- From (1) and (2) we have **Bayes formula**:

$$P(S|Q) = \frac{P(S)P(Q|S)}{P(Q|S)P(S) + P(Q|S^c)P(S^c)}$$

- Substituting what we know yields: $P(S|Q) \approx 4.5\%$

Bayes formula

- This means that **only approximately 4.5% of the people who take the test and got a positive result are actually sick.**
- The test seems to be **USELESS!**
- Why is this happening?? The test has “impressive credentials”: $P(Q|S) = 0.95$ and $P(Q|S^c) = 0.02$
- **The Reason:** the disease is rare.
 - **The disease is less likely to happen, $P(S) = 0.001$, than the failure of the test $P(Q|S^c) = 0.02$.**

Exercise

- Three friends A, B, and C run for president. Only one will win.
- We assume that A and B have equal chance of winning.
- On the other hand, C has $\frac{1}{2}$ the chance of A.
- Define the sample space Ω (recall, it is the set of possible outcomes).
- Compute the probabilities of each candidate becoming the president.
